

Fault Aware Training For Spiking Neuron Network

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Abstract—Spiking Neural Network (SNN) is gaining attention for their energy-efficient and biologically inspired computation, making them highly suitable for neuromorphic hardware. However, such hardware is vulnerable to faults, leading to performance degradation. This paper introduces a Fault Aware Training (FAT) framework to improve the robustness of SNN under hardware fault conditions. By simulating faults during training and employing targeted weight recovery, FAT ensures high fault tolerance while maintaining competitive classification accuracy. Experimental results demonstrate FAT’s effectiveness in enhancing SNN reliability on the MNIST dataset.

Index Terms—Spiking Neural Networks, Fault Tolerance, Neuromorphic Computation, Robust Training, Hardware Faults.

I. INTRODUCTION

With the rapid advancement of artificial intelligence and neuromorphic computing, Spiking Neural Networks (SNNs) have emerged as a promising computational model, mimicking the behavior of biological neural systems. Due to their event-driven nature and low power consumption, SNNs are increasingly favored for applications in energy-efficient computation and real-time processing. However, the practical implementation of SNNs often faces challenges from hardware faults and environmental uncertainties, such as neuron failures, synaptic connection errors, or random noise, which can significantly degrade network performance and reliability. Fault Training has been introduced as an effective approach to enhance the fault tolerance of SNNs during the training process. We believe that in many applications, SNNs cannot afford errors in image recognition, making the study of fault tolerance an essential topic.

In our research, we drew inspiration from the methodology proposed by Zahid et al. [1] in their paper, "FAT: Training Neural Networks for Reliable Inference Under Hardware Faults.". Their work explores training neural networks to maintain reliable inference under hardware faults, which provided critical guidance for designing our fault training methodology.

II. BACKGROUND

Spiking Neural Network (SNNs) is a bio-inspired model of computation that mimics the behavior of biological neurons. Unlike traditional artificial neural networks, SNN processes information using discrete spikes, making them highly energy-efficient and suitable for neuromorphic hardware. However, the deployment of SNNs on physical devices introduces challenges such as hardware-induced faults caused by environmental factors, manufacturing defects, or prolonged use.

Hardware faults in neuromorphic systems can lead to significant performance degradation, as errors in neuron activations or synaptic weights propagate through the network. Traditional fault-tolerant mechanisms, such as redundancy or error correction codes, often come with additional computational and resource costs, which undermine the efficiency of neuromorphic systems.

To train SNNs effectively, backpropagation[1] is a key method used to optimize their parameters. Unlike conventional neural networks, where gradients can be directly computed, SNN faces unique challenges due to their discrete spike-based activations. To address this, surrogate gradient methods are commonly employed, which approximate the non-differentiable spike function with a smooth and differentiable surrogate function. This allows the gradients to be propagated through the network layers, enabling efficient parameter updating. Additionally, techniques such as gradient clipping are used to stabilize training and prevent exploding gradients, particularly in deep SNN architectures.

In recent years, there has been increasing interest in developing software-level methods to enhance the fault tolerance of SNN. These methods aim to integrate fault resilience directly into the training process, reducing the reliance on hardware-level solutions. This paper builds on this trend by introducing a novel fault-aware training (FAT) framework, which simulates fault conditions during training to improve the robustness of SNN.

III. PROPOSED TECHNIQUE

The proposed Fault Awareness Training (FAT) framework is designed to enhance the fault tolerance of spiking neural networks (SNN) deployed on neuromorphic hardware. The framework incorporates fault simulation and adaptive training strategies to improve the robustness of SNN under various fault conditions.

A. Fault Simulation

Fault simulation is a core component of the FAT framework. During training, faults are introduced into the model to mimic real-world hardware failures, such as neuron activation errors or stuck@ weight corruptions. This is achieved using the `gen_faultlist()` function, which generates a fault list based on predefined probabilities and fault models, such as the "Hard to Spike Fault". We inject neuron fault to each neuron with a probability 0.25, if the neuron is injected fault, we let it's

weight update a wrong value randomly selected between -10 to 10. By injecting these faults during training, the SNN learns to adapt and maintain performance despite the presence of errors.

B. Weight Recovery

To counteract the adverse effects of injected faults, FAT employs a targeted weight recovery mechanism. During each training iteration, 25% neurons are randomly selected and their weights are restored to their original values after backpropagation. This selective restoration prevents the network from overfitting to fault-induced distortions and stabilizes the training process. Additionally, this approach ensures that critical weights are less likely to be permanently corrupted by faults.

C. Gradient Clipping

To further enhance the stability of the training, FAT integrates gradient clipping, limiting the maximum gradient norm during backpropagation. This prevents exploding gradients, which are particularly problematic in deep SNN architectures with spike-based activations. By capping the gradient norm to a threshold of 10, the training process remains stable and convergent.

D. Evaluation Under Fault-Free and Faulty Conditions

After training, the FAT framework evaluates the SNN under both fault-free and faulty conditions. This involves using the `good_inference()` function for fault-free scenarios and the `faulty_inference()` function for faulty scenarios.

To visualize the impact of FAT, the accuracy of the SNN under fault-free and faulty conditions is plotted. In addition, a box plot is used to statistically compare the two scenarios, demonstrating the robustness of the FAT-trained network. These visualizations provide insights into the effectiveness of the proposed method in mitigating hardware faults.

E. Key Contributions

The proposed FAT framework offers the following contributions:

- Integration of fault simulation during training to improve SNN robustness.
- Introduction of targeted weight restoration to stabilize training under fault conditions.
- Demonstration of FAT's effectiveness through accuracy comparisons and statistical analysis.

Experimental results confirm that FAT significantly enhances fault tolerance, making it a promising approach for deploying SNN on fault-prone neuromorphic hardware.

IV. CONCLUSION

In this study, we evaluated the robustness of normal backpropagation (BP) and modified backpropagation (BP) techniques for Spiking Neural Network (SNN) under fault-free and faulty conditions. The experimental results, as shown in Table I and Figure 1, demonstrate that:

1. Modified BP consistently outperforms normal BP in both fault-free and faulty scenarios.
2. The modified BP approach achieves higher accuracy under faulty conditions, highlighting its robustness in handling hardware faults.
3. The average accuracy improvement under faulty conditions is approximately 2.0%.

These findings validate the effectiveness of integrating fault-aware mechanisms in training strategies for neuromorphic systems.

TABLE I
ACCURACY COMPARISON BETWEEN NORMAL AND MODIFIED BACKPROPAGATION

Experiment	Normal BP Fault-Free	Normal BP Faulty	Modified BP Fault-Free	Modified BP Faulty
1	82.34	78.20	82.38	79.43
2	81.59	76.09	82.82	79.16
3	82.30	77.48	82.93	79.50
4	82.11	77.11	82.83	79.13
5	82.04	77.16	81.81	78.12
6	82.14	76.15	82.43	78.97
7	82.42	76.45	82.57	78.69
8	82.19	77.32	82.02	78.77
9	81.47	77.39	82.74	78.77
10	81.64	76.86	83.53	79.82

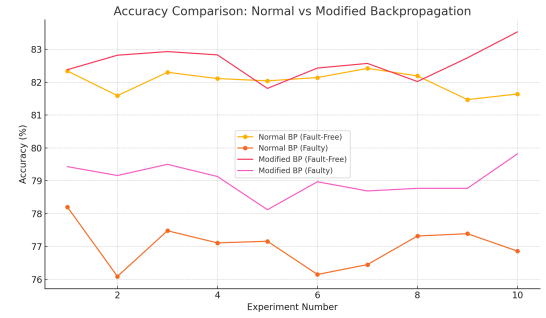


Fig. 1. Accuracy Comparison Between Normal and Modified Backpropagation

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